

GradeFrame | Bug Fix Log

BITS Digital CodeForge V1.0 - Local verified build - 27 September 2026

#	Bug / issue identified	How reproduced	Root cause	Fix implemented	How tested
B01	Reset fails before course selection	Open the original app and confirm both Reset Range prompts before selecting a course. Browser harness reproduces the exception.	The handler dereferences grade controls before buildGradeUI has created them.	Disable Reset until a course has records; also guard the handler. Provide Undo instead of two confirmation prompts.	Browser regression: reset is disabled on initial load; reset after invalid edits restores valid boundaries and export.
B02	Duplicate course options within one upload	Import the boundary fixture: four records belonging to Course A. The original creates four Course A options.	The upload handler adds an option for every record without deduplication.	Build the course list from a Set of validated course names.	Browser regression: the same four-record XLSX produces exactly one course option.
B03	Old courses survive replacement uploads	Import the Course A fixture, then the one-record Course B fixture. Original options include four Course A entries plus Course B.	The upload handler appends options and replaces data without resetting dependent UI state.	A successful import atomically replaces data, options, ranges, history, filters and session timing. Failed imports retain the previous valid workbook with an explicit notice.	Browser regression: replacement leaves only Course B and updates its average to 65. Invalid replacement preserves the previous four valid records.
B04	Minimum and maximum displayed under swapped labels	Import marks 0, 20, 80 and 100. Original browser output reads Min 100 and Max 0.	The Min label contains the max element ID; the Max label contains the min element ID.	Render correctly named minimum and maximum fields from the shared statistics function.	Browser regression: Min 0, Max 100, mean 50 and median 50. Core tests independently check the expected values.

Evidence: the original app was executed unchanged in a same-origin browser iframe with synthetic XLSX files. The repaired app passed 50 browser assertions and 15 core/report tests, including exhaustive integer-boundary checks.

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B05	Incomplete grade coverage permits export	Set A maximum to 99, or E minimum to 1, in the original. Validation reports no error and export stays enabled even with boundary-mark students.	Validation checks adjacent bands but does not require the outer endpoints to be 100 and 0.	Validate complete 0-100 coverage in the core and at export time; lock the outer endpoint controls in the UI.	Core tests reject either missing endpoint, gaps and overlaps. Browser regression verifies locked endpoints and blocked invalid exports. All 101 integer marks are checked across defaults and 500 valid partitions.
B06	Valid single-mark grade band is rejected	Set A to 100-100 and A- maximum to 99. Original validation rejects the otherwise continuous ranges.	The condition $\min \geq \max$ disallows equality, although ranges are inclusive and marks are whole numbers.	Allow equal minimum and maximum while preserving continuity, ordering and endpoint checks.	Core tests assign 100 to A and 99 to A-. Browser regression accepts the single-mark band and updates its adjacent maximum.
B07	Empty course displays NaN statistics	After importing marks, choose the blank course option in the original. Mean and median become NaN, and export remains enabled with valid ranges.	Statistics divide by zero and index an empty array; export enablement depends only on range validity.	Return explicit null statistics for empty data and render dashes. Require records, valid ranges and a nonblank instructor for export.	Core empty-data test and initial browser empty-state check pass. Browser checks verify export blocking with no instructor and after clearing the instructor.
B08	File picker advertises only .xls despite .xlsx requirement	Inspect the original file input: accept is .xls. The challenge specifies .xlsx workbooks.	The HTML accept attribute names only the legacy extension.	Accept .xlsx and .xls, validate the file signature and parse the workbook with a pinned local Excel reader.	Native file selection imports the four-record boundary XLSX. Browser regression rejects corrupt input. The actual downloaded CSV was reopened and all four expected IDs, marks and grades were verified.

End-to-end verification: the native file chooser imported a real XLSX fixture. The actual saved CSV was reopened independently and verified against all four expected records. Desktop and mobile layouts were inspected. New report native saving and public deployment testing remain pending; see verification.md.